

HASSAN MAHMOOD

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EDUCATION

Northeastern University

Ph.D. Computer Science

Advisor: Dr. Ehsan Elhamifar

Research Interests: Generative Models, Adversarial Attacks and Robustness, Computer Vision

Boston, MA

Aug. 2020 - Present

National University of Sciences and Technology

Bachelor of Software Engineering

Advisor: Dr. Faisal Shafait

Research Interests: Computer Vision, Real-time Vehicle Detection and Localization

Islamabad, Pakistan

Sep. 2014 – July 2018

PUBLICATIONS

- **Hassan Mahmood**, Ehsan Elhamifar, “Compositional Targeted Multi-Label Universal Perturbations,” **CVPR 2025**. [PDF]
Spotlight presentation — [Uncertainty Quantification for Computer Vision Workshop \(CVPR 2025\)](#).
- **Hassan Mahmood**, Ehsan Elhamifar, “Semantic-Aware Multi-Label Adversarial Attacks,” **CVPR 2024**. [PDF]
- S. R. Qasim, **Hassan Mahmood**, Faisal Shafait, “Rethinking Table Recognition using Graph Neural Networks,” **ICDAR 2019**. [PDF]

WORK EXPERIENCE

Graduate Research Assistant

MCADS Lab, Northeastern University

Aug. 2020 – Present

Boston, MA

- Working on **diffusion** generative models to meta-learn training data modifications to improve score estimation for low density samples. The goal is to find adversarial modifications that are captured well through normal diffusion training.
- Worked on **Diffusion Classifiers** for faster inference and evaluated their robustness to adversarial and distribution shifts.
- Proposed a knowledge graph-based framework for **multi-label adversarial attacks**, which addressed the negative gradient correlation problem to enhance attack efficiency. This optimization significantly accelerated attack speed while improving attack success rates by at least **20%** across benchmark datasets like PASCAL-VOC, NUS-WIDE, and OpenImages.
- Designed an optimization framework to generate scalable **targeted universal perturbations** for multi-label learning models, reducing the computational complexity from **exponential** to **linear** while maintaining comparable attack performance. We presented empirical evidence suggesting approximate linearity of universal adversarial regions of deep learning models.
- Improved text-based error generation for procedural task videos by prompt tuning Llama-2 and using adversarial methods with iterative feedback from a secondary Llama-2, yielding a **35%** increase in valid error generation over the baseline.

Graduate Research Assistant

CBL Lab, University of Houston

Aug. 2019 – July 2020

Houston, TX

- Developed **interpretable and explainable** deep learning models for cardiovascular disease prediction, surpassing benchmarks with a **6%** improvement in F1-score on the MESA dataset.
- Proposed a **privacy-preserving affective computing** solution using gait recognition and temporal modeling to analyze customer behavior, contributing to securing a **\$100K** in research grant.

AI Researcher

Deep Learning Lab, National Center of Artificial Intelligence

July 2018 – Aug. 2019

Islamabad, Pakistan

- Proposed a **Graph Neural Network (GNN)**-based approach for recognizing table structures in document images, achieving a **37%** improvement in accuracy over baselines.
- Developed an algorithm to generate a synthetic dataset of tabular data images with complex layouts. [Link]

Other Machine Learning Projects

- Collaborated to design an outfit classification and recommendation system for (**Trendage**), a US-based startup.
- Authored a comprehensive e-book tutorial on machine learning and deep learning for **AI Sciences**.
- Implemented a deep learning-based character recognition pipeline for LCD queue displays, contributing to a grant proposal of **Talents Fellowship Program** at the Machine Learning Lab, University of Trieste.

ENGINEERING SKILLS

- **Programming Languages:** Python, C/C++, Java, SQL, Javascript, Matlab
- **ML Frameworks/Tools:** Pytorch, Keras, Tensorflow, Jupyter, OpenCV, Numpy, Scipy
- **Others:** Git, AWS, Selenium, Flask, Wireshark, TCP/IP Protocol, DNS Protocol

SERVICES

- Served as a reviewer for TPAMI journal, CVPR, AAAI, ICCV, and ICLR conferences.
- Supervised an undergraduate research intern for six months at University of Houston.

AWARDS

- Selected as one of the 190 students from a pool of 2500+ candidates across Pakistan for fully funded UGRAD semester exchange program sponsored by the U.S. Department of State (worth $\approx$ **\$22,500**).
- Prime Minister National ICT R&D Merit Scholarship for fully funded undergraduate degree and monthly stipend for four years (worth $\approx$ **\$12,000**).
- Bronze medal and merit scholarship for 3rd standing (in a total of $\approx$ **40,000** students) in computer science higher education exams in Pakistan - FBISE, Islamabad.